🛡️ Wazuh SIEM Home Lab

📌 Project Description

This project demonstrates the deployment and use of **Wazuh SIEM** in a virtual laboratory environment for centralized security monitoring, endpoint visibility, and basic threat detection.

The lab is designed to demonstrate the following activities:

✅ Deploying Wazuh Manager, Indexer, and Dashboard.
✅ Connecting a Linux endpoint using the Wazuh agent.
✅ Configuring **File Integrity Monitoring (FIM)**.
✅ Monitoring failed SSH authentication attempts.
✅ Simulating SSH brute-force activity from Kali Linux using Hydra.
✅ Reviewing security alerts, rule IDs, severity levels, and MITRE ATT&CK mappings where available.
✅ Documenting investigation findings and incident response recommendations.

The purpose of this project is to gain practical experience with tasks commonly performed by a **Security Operations Centre (SOC) Analyst**.

> **Lab safety:** Perform all testing only against systems you own or are explicitly authorized to test. Keep the lab isolated and do not target third-party systems.

🔧 Tools Used

Tool	Description
Wazuh SIEM	Open-source security monitoring platform used to collect and analyze endpoint security events.
Wazuh Agent	Endpoint component that forwards relevant events and monitoring data to the Wazuh server.
Ubuntu Server	Operating system used for the Wazuh server and monitored Linux endpoint.
Kali Linux	Security testing machine used for authorized lab simulations.
Oracle VirtualBox	Virtualization software used to create and manage the lab machines.
Hydra	Tool used to generate controlled SSH authentication attempts in the lab.
SSH	Protocol used for remote access and authentication testing.

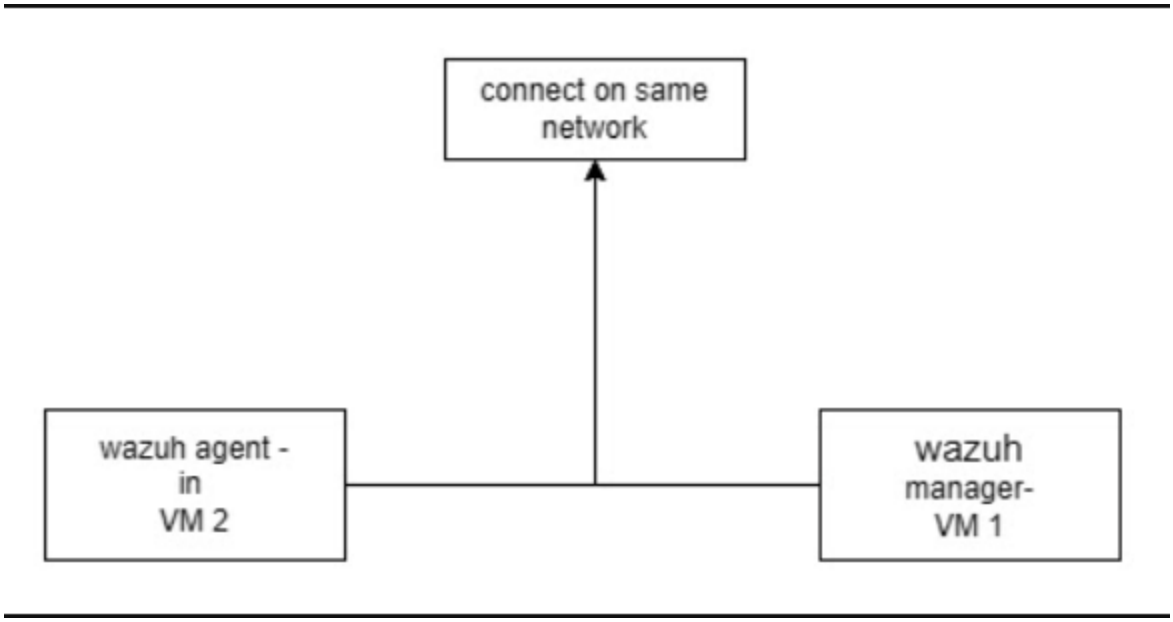
🏗️ Lab Architecture

The lab contains three virtual machines:

Virtual Machine	Operating System	Role	RAM	Storage
`wazuh-server`	Ubuntu Server	Wazuh Manager, Indexer, and Dashboard	4 GB	40GB
`agent-vm`	Ubuntu Server	Monitored endpoint running the Wazuh agent	2 GB	20GB
`kali-attacker`	Kali Linux	Authorized security testing machine	2 GB	20GB

Architecture Diagram

Add your lab diagram here:



Suggested flow: Kali Linux (testing machine) → Linux endpoint (Wazuh agent) → Wazuh server → Wazuh Dashboard.

🛠️ Installation & Setup

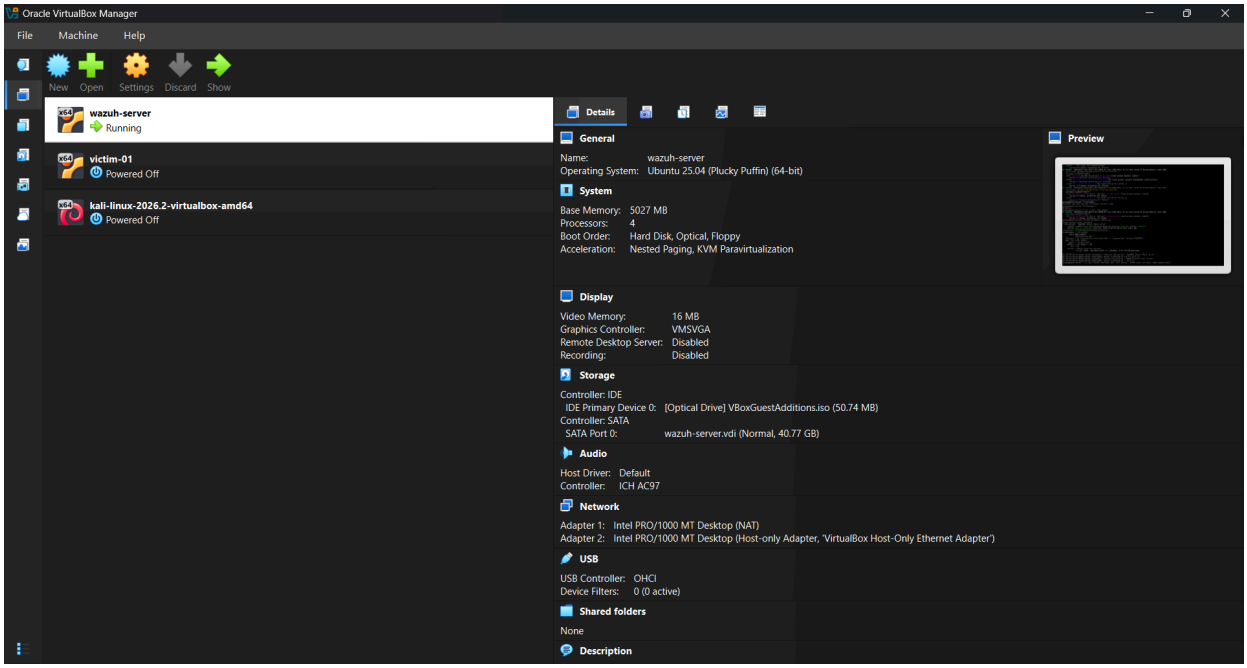
1️⃣ Create the Virtual Machines

Create the required virtual machines in Oracle VirtualBox:

- `wazuh-server` — Ubuntu Server.
- `agent-vm` — Ubuntu Server.
- `kali-attacker` — Kali Linux.

Configure CPU, memory, storage, and networking according to the resources available on your computer.

Evidence:



2️⃣ Install Wazuh SIEM

Follow the official Wazuh installation documentation:

🔗 [Wazuh Installation Guide]

Update the Ubuntu server:

```
Shell
sudo apt update && sudo apt upgrade -y
```

Download the installation assistant:

```
Shell
curl -sO https://packages.wazuh.com/4.x/wazuh-install.sh
```

Run the all-in-one installation:

```
Shell
sudo bash ./wazuh-install.sh -a
```

The all-in-one installation deploys the Wazuh Manager, Indexer, and Dashboard on the server.

Important: The installation command and package version should be checked against the current official Wazuh documentation before use.

3️⃣ Access the Wazuh Dashboard

Open a browser on the host computer and visit:

```
None
https://<wazuh-server-ip>
```

Sign in using the credentials generated during installation.

- **Username:** `admin`
- **Password:** Retrieve the generated password from the installation output or password file. Do not publish it.

4️⃣ Deploy and Connect the Linux Agent

In the Wazuh Dashboard:

1. Open **Agents**.
2. Select **Deploy new agent**.
3. Choose the appropriate Linux package and architecture.
4. Enter the Wazuh server address.
5. Follow the generated installation and enrollment commands on `agent-vm`.

Enable and start the agent:

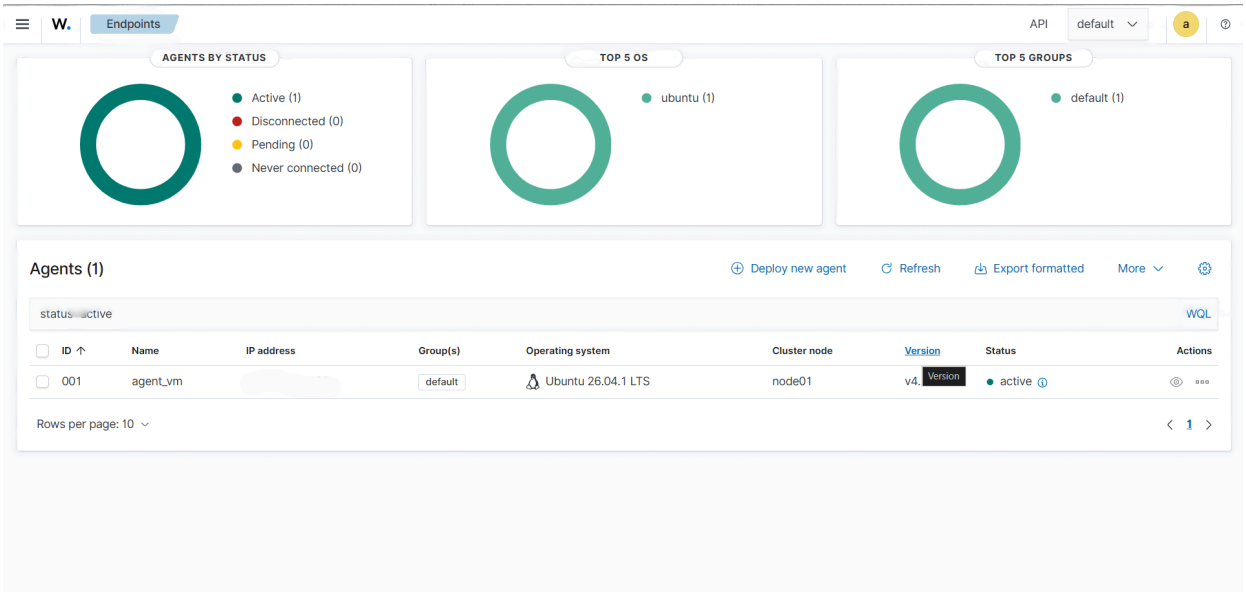
```
Shell
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```

Verify the service:

```
Shell
sudo systemctl status wazuh-agent
```

Confirm that the agent appears as **Active** in the Wazuh Dashboard.

Evidence:



🔍 Security Monitoring & Detection Tests

5️⃣ File Integrity Monitoring (FIM)

Objective: Monitor a selected directory and identify file creation, modification, and deletion events.

Create a test directory on the monitored endpoint:

```
Shell
mkdir ~/secure-folder
```

Edit the Wazuh agent configuration:

```
Shell
sudo nano /var/ossec/etc/ossec.conf
```

Add the directory inside the appropriate `<syscheck>` section, replacing `youruser` with the actual Linux username:

```xml
<directories realtime="yes">/home/youruser/secure-folder</directories>
```

Restart the agent:

```shell
sudo systemctl restart wazuh-agent
```

Perform controlled file tests:

```shell
echo "test file" > ~/secure-folder/test-file.txt
```

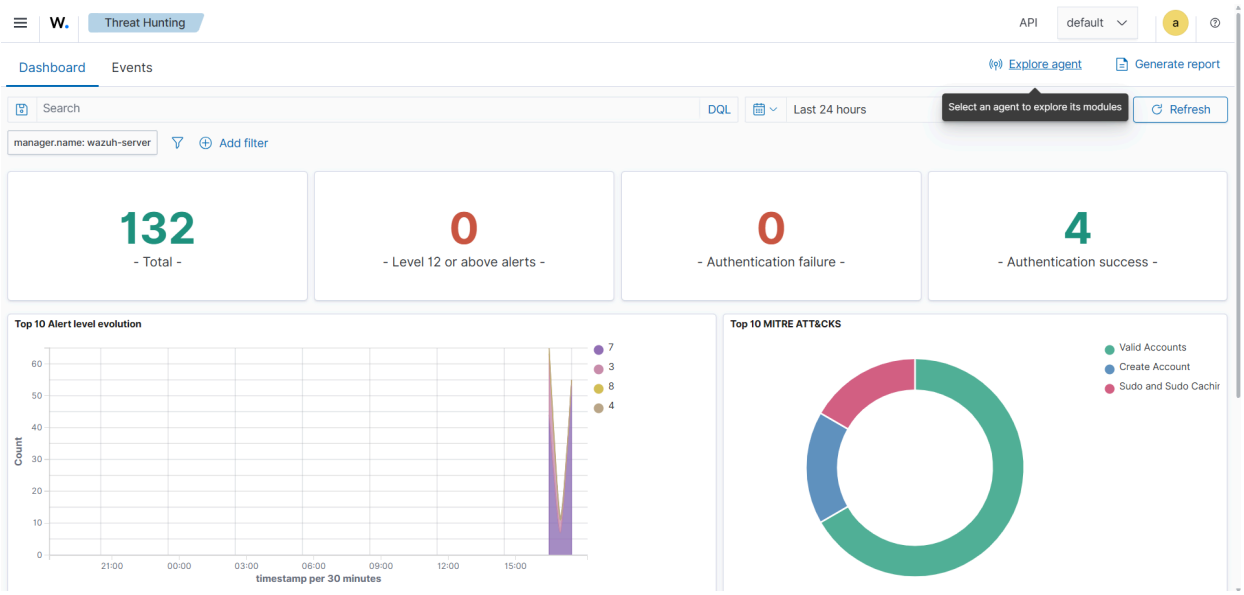
Modify the file, then delete it:

```shell
echo "updated content" >> ~/secure-folder/test-file.txt
rm ~/secure-folder/test-file.txt
```

Review the resulting events in the Wazuh Dashboard under the relevant File Integrity Monitoring or Threat Hunting views.

Record your actual observations:

Evidence:



6️⃣ Failed SSH Authentication Monitoring

Objective: Observe how failed SSH authentication attempts are recorded and surfaced by Wazuh.

From an authorized lab machine, attempt to connect to the monitored endpoint using an intentionally incorrect username or password:

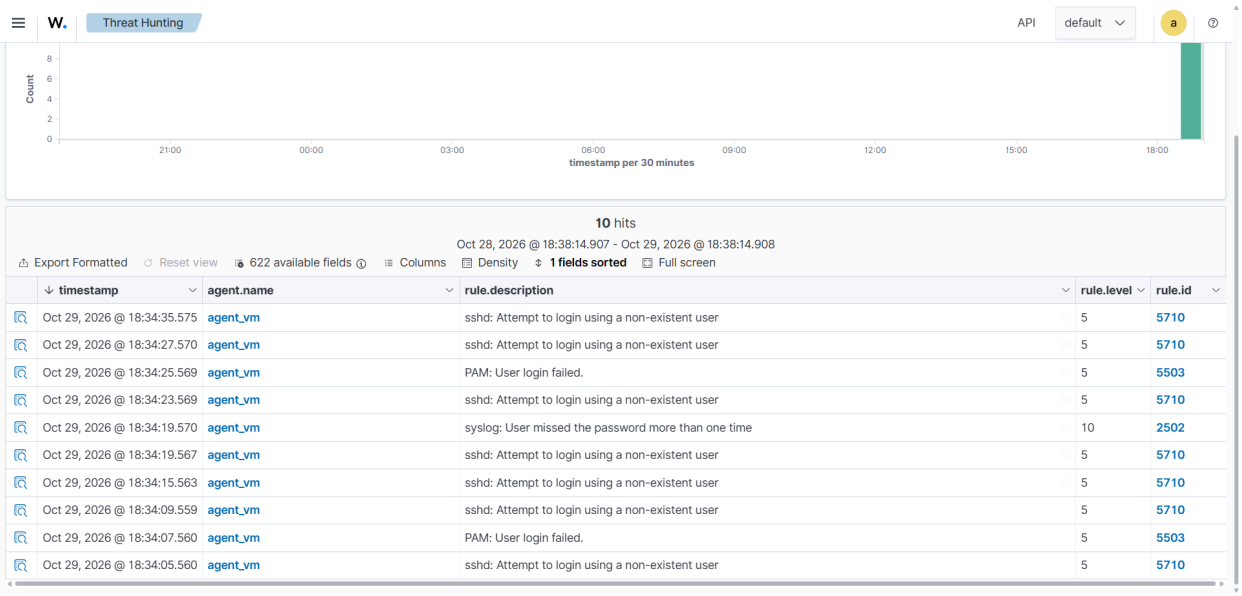
```
Shell
ssh wronguser@<agent-vm-ip>
```

Review the endpoint authentication events in Wazuh Threat Hunting.

Record your observations:

- **Test date/time:** [Enter value]
- **Target endpoint:** [Enter lab endpoint name]
- **Detection status:** [Detected / Not detected]
- **Alert description:** [Enter alert description]
- **Rule ID:** [Enter rule ID]
- **Severity:** [Enter severity level]

Evidence:



timestamp	agent.name	rule.description	rule.level	rule.id
Oct 29, 2026 @ 18:34:35.575	agent_vm	sshd: Attempt to login using a non-existent user	5	5710
Oct 29, 2026 @ 18:34:27.570	agent_vm	sshd: Attempt to login using a non-existent user	5	5710
Oct 29, 2026 @ 18:34:25.569	agent_vm	PAM: User login failed.	5	5503
Oct 29, 2026 @ 18:34:23.569	agent_vm	sshd: Attempt to login using a non-existent user	5	5710
Oct 29, 2026 @ 18:34:19.570	agent_vm	syslog: User missed the password more than one time	10	2502
Oct 29, 2026 @ 18:34:19.567	agent_vm	sshd: Attempt to login using a non-existent user	5	5710
Oct 29, 2026 @ 18:34:15.563	agent_vm	sshd: Attempt to login using a non-existent user	5	5710
Oct 29, 2026 @ 18:34:09.559	agent_vm	sshd: Attempt to login using a non-existent user	5	5710
Oct 29, 2026 @ 18:34:07.560	agent_vm	PAM: User login failed.	5	5503
Oct 29, 2026 @ 18:34:05.560	agent_vm	sshd: Attempt to login using a non-existent user	5	5710

7️⃣ SSH Brute-Force Simulation Using Hydra

Objective: Generate repeated SSH authentication attempts against the lab endpoint and examine whether Wazuh identifies suspicious activity.

Testing environment:

- **Testing machine:** Kali Linux.
- **Target:** Ubuntu Linux endpoint running the Wazuh agent.
- **Monitoring platform:** Wazuh SIEM.
- **Testing tool:** Hydra.

Create a small test password list containing dummy values, then run the test only against your own lab endpoint.

Example command:

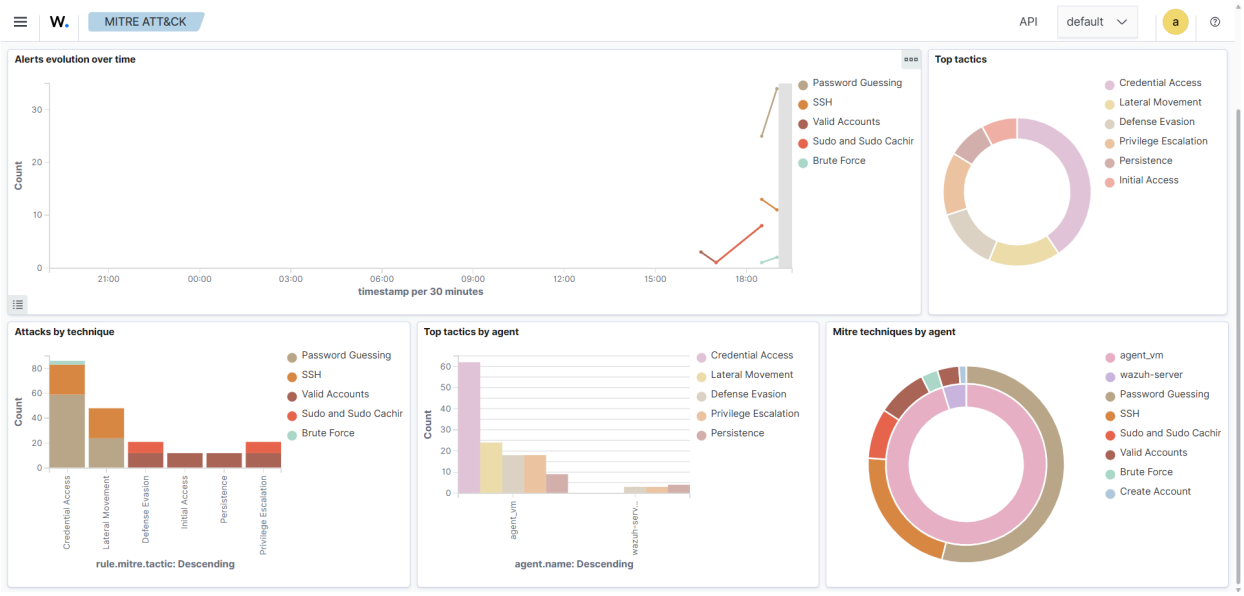
```
Shell
hydra -l youruser -P pass.txt ssh://<agent-vm-ip> -t 4
```

Review the generated authentication events in the Wazuh Dashboard.

Investigate the available alert details, including:

- Alert description.
- Rule ID.
- Severity level.
- Source IP address.
- Timestamp.
- MITRE ATT&CK technique or tactic, if provided.

Evidence:



📊 Alert Analysis & Investigation

Use this table to summarize the results of your tests. Fill in the values from your own Wazuh alerts.

Example Investigation Notes

For each important alert, document:

1. **What happened?** Describe the activity that generated the event.
2. **Why is it relevant?** Explain why the activity may be suspicious.
3. **What evidence is available?** Record the timestamp, source, rule ID, severity, and relevant log details.
4. **What is your assessment?** Distinguish expected lab activity from suspicious activity. Do not label an event a confirmed incident without supporting evidence.
5. **What should happen next?** Describe suitable containment, remediation, or monitoring actions.

🚨 Incident Response

Investigation Workflow

For a suspicious SSH authentication alert, use the following investigation approach:

1. **Identify the alert:** Locate the relevant event in Wazuh.
2. **Review the source:** Examine the source IP and affected endpoint.
3. **Review authentication logs:** Check whether any authentication attempt succeeded.
4. **Assess the activity:** Determine whether the pattern is consistent with repeated unauthorized login attempts.
5. **Recommend response actions:** Consider blocking the source where appropriate, enforcing SSH key-based authentication, disabling password authentication where suitable, and applying rate limiting.
6. **Verify the outcome:** Check whether the activity stopped and document the evidence.

📌 Results & Key Findings

Summarize the results from your completed tests.

Key Findings

- **Centralized monitoring**
- **File integrity monitoring**
- **Authentication monitoring**
- **Brute-force simulation**
- **Alert investigation**

🎯 Skills Demonstrated

This project provided hands-on exposure to:

- SIEM deployment and configuration.
- Wazuh Manager, Indexer, Dashboard, and Agent concepts.
- Linux system administration.
- Endpoint log collection and monitoring.
- File Integrity Monitoring.
- SSH authentication log analysis.

- Security alert triage and investigation.
- Rule IDs and alert severity.
- MITRE ATT&CK mapping, where available.
- Basic incident response procedures.
- Virtual machine and lab network configuration.

🚀 Future Enhancements

Potential improvements for this lab include:

- 🔹 Add a Windows endpoint and monitor Windows security events.
- 🔹 Create and test custom Wazuh detection rules.
- 🔹 Configure Wazuh Active Response and verify its behavior.
- 🔹 Explore vulnerability detection and Security Configuration Assessment.
- 🔹 Integrate threat intelligence services such as VirusTotal.
- 🔹 Configure notifications for selected security alerts.
- 🔹 Expand the lab with additional monitored endpoints.

📚 References

- Wazuh Official Documentation
- Wazuh Installation Guide
- MITRE ATT&CK Framework
- Kali Linux Documentation
- Oracle VirtualBox Documentation

📬 Contact

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This is a hands-on cybersecurity home lab project created for learning and demonstrating practical SIEM monitoring and SOC investigation skills.